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For the molecule to escape, total energy at Earth's surface must at least be zero (min energy at infinity).

$$\text{Hence, } -\frac{GMm}{r} + \frac{1}{2}mv^2 = 0 \rightarrow v = \sqrt{\frac{2GM}{R_E}}$$

Therefore,

$$v_{\text{new}} = \sqrt{\frac{2GM}{3R_E}} = \frac{1}{\sqrt{3}} \sqrt{\frac{2GM}{R_E}} = \frac{1}{\sqrt{3}} \times 1.1 \times 10^4 \\ = 6.4 \times 10^3 \text{ m s}^{-1}$$

Alternatively, use ratio method

$$v \propto \sqrt{\frac{1}{r}}, \text{ therefore } \frac{v_{\text{new}}}{v_{\text{original}}} = \frac{r_{\text{original}}}{r_{\text{new}}} \\ \frac{v_{\text{new}}}{1.1 \times 10^4} = \sqrt{\frac{R_E}{3R_E}} \\ v_{\text{new}} = \sqrt{\frac{1}{3}} (1.1 \times 10^4) \\ = 6.4 \times 10^3 \text{ m s}^{-1}$$

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